

SARXarray: Xarray extension for Synthetic Aperture Radar data

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Software

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Summary

Satellite-based Synthetic Aperture Radar (SAR) provides invaluable data for Earth Observation. The Interferometric SAR (InSAR) technique ([Hanssen, 2001](#)), which utilizes a stack of SAR images in Single Look Complex (SLC) format, plays a significant role in various surface motion monitoring applications, e.g. civil-infrastructure stability ([Chang et al., 2017](#); [Chang & Hanssen, 2014](#); [Özer et al., 2018](#)), and hydrocarbons extraction ([Fokker et al., 2016](#); [Zhang et al., 2022](#)). To facilitate advanced data processing for InSAR communities, we developed SARXarray, a Xarray extension for handling co-registered SLC SAR stacks.

Statement of Need

Satellite-based SAR generates data stacks with long temporal coverage, wide spatial coverage, and high spatio-temporal resolution ([Moreira et al., 2013](#)). Handling SAR data stacks in an efficient way is a common challenge within the InSAR community. To address this challenge, High-Performance Computing (HPC) is often used to process data in a parallel and distributed manner. However, to fully leverage HPC capabilities, data processing workflows need to be customized for each specific use-case.

To facilitate efficient processing of SLC SAR stacks and minimize code customization, we developed SARXarray.

SARXarray leverages two well-established Python libraries Xarray and Dask from the [Pangeo community](#). It utilizes Xarray's support on labeled multi-dimensional datasets to stress the space-time character of an SLC SAR stack. Dask is used to perform lazy evaluation of operations and block-wise computations. SARXarray can be integrated into existing Python workflows of InSAR processing and deployed on a variety of computational infrastructures.

State of the field

A similar open-source library [xarray-sentinel](#) exists for handling raw Sentinel-1 GRD and SLC data as lazy-loaded Xarray Datasets. Despite the similar goals of digesting SAR data into lazy-loaded Xarray Datasets, SARXarray and [xarray-sentinel](#) are designed for different applications:

- SARXarray is designed to handle co-registered SLC stacks, instead of raw Sentinel-1 data. It supports the output of two common SLC co-registration tools: [DORIS](#) and [SNAP](#), instead of raw Sentinel-1 data products from European Space Agency (ESA).
- SARXarray supports interferometric stacks from other sensors than Sentinel-1, as long as they can be co-registered by SNAP or DORIS. It reads the output from the two co-registration tools, and relies on them to handle the specificities of different sensors.

Software design

SARXarray is designed as an extension of Xarray using accessors. This design is motivated by [Xarray community's recommendation](#), in order to isolate the extension from API changes of the core Xarray library. The software has three main components: an I/O module that lazily loads binary SLCs and related metadata, a Stack accessor that provides SAR-specific operations (amplitude/phase extraction, multi-looking, Mean-Reflection-Map generation, coherence calculation), and utility functions.

Research impact statement

SARXarray is a dependency of [CAROLINE](#) (Contextual and Autonomous processing of satellite Radar Observations for Learning and Interpreting the Natural and built Environment), which is an InSAR processing framework developed by the InSAR group of Delft University of Technology. It has facilitated the data processing of multiple InSAR research projects ([Brouwer & Hanssen, 2025](#); [Conroy et al., 2024](#); [Lumban-Gaol et al., 2025](#)).

SARXarray is a recognized related project of the Xarray ecosystem, see the [Xarray user-guide page](#).

Tutorial

We provide a tutorial as a Jupyter notebook to demonstrate the functionalities of SARXarray:

[Tutorial Jupyter notebook](#)

The tutorial includes the following steps:

- Installation and data preparation
- Lazy loading a SAR data stack in binary format as an Xarray Dataset
- Attaching attributes to the loaded stack
- Applying common SAR operations on the loaded stack such as:
 - Multi-Looking
 - Creation of a Mean-Reflectivity-Map (MRM)
 - Calculation of coherence

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AI usage disclosure

In writing the software documentation and the JOSS paper, GPT-5 was used for language improvements for grammar correction and style edits.

Additionally, various language models were used via [GitHub Copilot](#) in Pull Requests reviews, for pre-filtering obvious issues such as typos, small logical errors, and for suggesting code improvements. All suggestions from AI were reviewed and verified by the authors before merging into the codebase. The correctness of the code was ensured by the unit tests.

References

- Brouwer, W. S., & Hanssen, R. F. (2025). On the definition of an independent stochastic model for InSAR time series. *IEEE Transactions on Geoscience and Remote Sensing*, 63, 1–11. <https://doi.org/10.1109/TGRS.2025.3600893>
- Chang, L., Dollevoet, R. P. B. J., & Hanssen, R. F. (2017). Nationwide railway monitoring using satellite SAR interferometry. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 10(2), 596–604. <https://doi.org/10.1109/JSTARS.2016.2584783>
- Chang, L., & Hanssen, R. F. (2014). Detection of cavity migration and sinkhole risk using radar interferometric time series. *Remote Sensing of Environment*, 147, 56–64. <https://doi.org/10.1016/j.rse.2014.03.002>
- Conroy, P., Lumban-Gaol, Y., Diepen, S. van, Leijen, F. van, & Hanssen, R. F. (2024). First wide-area dutch peatland subsidence estimates based on InSAR. *IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium*, 10732–10735. <https://doi.org/10.1109/IGARSS53475.2024.10642504>
- Fokker, P., Wassing, B., van Leijen, F., Hanssen, R., & Nieuwland, D. (2016). Application of an ensemble smoother with multiple data assimilation to the bergermeer gas field, using PS-InSAR. *Geomechanics for Energy and the Environment*, 5(March), 16–28. <https://doi.org/10.1016/j.gete.2015.11.003>
- Hanssen, R. F. (2001). *Radar interferometry: Data interpretation and error analysis*. Kluwer Academic Publishers. <https://doi.org/10.1007/0-306-47633-9>
- Lumban-Gaol, Y., Conroy, P., van Diepen, S., van Leijen, F., & Hanssen, R. (2025). Peat subsidence and dynamics in midden-delfland, the netherlands, from time series InSAR analysis and the SPAMS model. *Geoderma*, 463, 117551. <https://doi.org/10.1016/j.geoderma.2025.117551>
- Moreira, A., Prats-Iraola, P., Younis, M., Krieger, G., Hajnsek, I., & Papathanassiou, K. P. (2013). A tutorial on synthetic aperture radar. *IEEE Geoscience and Remote Sensing Magazine*, 1(1), 6–43.
- Özer, I. E., Leijen, F. J. van, Jonkman, S. N., & Hanssen, R. F. (2018). Applicability of satellite radar imaging to monitor the conditions of levees. *Journal of Flood Risk Management*, e12509.
- Zhang, B., Chang, L., & Stein, A. (2022). A model-backfeed deformation estimation method for revealing 20-year surface dynamics of the groningen gas field using multi-platform SAR imagery. *International Journal of Applied Earth Observation and Geoinformation*, 111, 102847. <https://doi.org/10.1016/j.jag.2022.102847>